

CS 362 / 562 - Assignment 1

Awad distinguishes between different “types” of AI. What classification scheme does the paper use and why do these types matter for scientific research?

In the paper, Awad identifies seven different types of AI schemes used in scientific research. Here is the list of those different types of AI schemes:

Predictive AI refers to methods that use existing data to anticipate outcomes, identify patterns, or simulate behaviors in both digital and physical systems.

Descriptive AI focuses on analyzing and interpreting large, complex, and often multimodal datasets.

Generative AI produces new content based on learned information, including text, images, computer code, or synthetic datasets.

Optimization AI is used to automate and improve decision-making processes, particularly in experimental design and parameter tuning.

Privacy-aware AI addresses the need to protect sensitive or restricted data while still allowing scientific analysis, often through techniques such as federated learning or differential privacy.

Meta-scientific AI refers to systems designed to support or enhance the scientific process itself, for example by assisting with literature review, experimental planning, or evaluation of research trends.

These distinctions matter because each type of AI emphasizes different capabilities—such as prediction, analysis, generation, or process optimization—and therefore plays different roles in scientific discovery. As Awad argues, AI in science is best understood as a constellation of paradigms rather than a single tool, and recognizing these differences helps clarify both the opportunities and limitations of AI in research.

Does Awad make a clear distinction between AI as a tool and AI as a scientific collaborator? If so, what are the differences and what are some examples given to support the differences? Do these examples suggest a real shift in how science is conducted, or mostly an extension of existing methods?

In this article, Awad makes a clear distinction between AI as a tool and AI as a scientific collaborator. The main difference lies in whether AI simply assists researchers with specific tasks or actively contributes to generating ideas and shaping the research process. One example Awad provides of AI as a tool is the use of large language models (LLMs) to reduce the time spent on routine coding, thereby facilitating the development of computational experiments, especially for interdisciplinary researchers or those with limited programming proficiency. In contrast, Awad presents meta-scientific AI as an example of AI acting as a collaborator, as these systems can partially automate the scientific discovery pipeline. I believe these examples demonstrate both perspectives: on one hand, most current AI applications extend existing research methods, but on

the other hand, meta-scientific AI reflects a growing trend in which AI functions more like an agent that can make decisions and adapt its behaviour based on outcomes.

What are some limitations or risks of using AI in science? How do these relate to issues such as interpretability, bias, reproducibility, or theory formation?

In the article, Awad identifies several risks associated with the use of AI in scientific research. One major concern is the “black box” nature of many AI models, which makes it difficult to explain and justify their outputs. This limitation is closely related to interpretability, since scientific knowledge requires transparent reasoning and validation, while black-box models can produce results that are hard for researchers to interpret or verify. Another limitation involves bias in training data. Awad notes that NLP tools may perpetuate and reflect unintentional societal biases, including race and gender stereotypes. When the underlying data contain bias, it becomes challenging to eliminate those biases from the model’s predictions. Reproducibility is also a concern, illustrated by the example of the Galactica model, which was withdrawn shortly after its release due to fabricated citations and misleading claims. Finally, AI raises issues related to theory formation because it often derives insights from statistical patterns rather than theoretical understanding. As a result, explanations may become data-driven rather than theory-driven, and scientists may struggle to interpret AI-generated hypotheses within established scientific frameworks.

According to Awad’s arguments, is AI more likely to *accelerate* scientific discovery or to *reshape* the scientific method itself? Do you agree or disagree?

From Awad’s perspective, I believe that he thinks AI is to reshape the scientific method. From the article, he claims that Artificial intelligence is reshaping the scientific enterprise, not only as a computational assistant, but as a generative force in knowledge production. It now contributes across disciplines to hypothesis generation, pattern recognition and predictive modelling, extending traditional methods of theory and experimentation

From my point of view, I agree with Awad's points. I think AI does more than simply speed up scientific research; it is beginning to change how scientists generate ideas, test hypotheses, and interpret results. AI systems are increasingly capable of identifying patterns in large datasets, proposing new research directions, and assisting in experimental design, which means they are influencing parts of the scientific process that were traditionally driven only by human reasoning. While human scientists still play a central role in validating findings and providing theoretical understanding, AI is reshaping how knowledge is discovered and evaluated. For this reason, AI should be understood not only as a powerful research tool but also as a technology that is gradually transforming the practice of science itself.